

# AWS Setup for hngac-fpga

## Screen-by-Screen Guide

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### What This Guide Is For

This guide is for a solo AWS account with no IAM Identity Center / SSO. It walks through the exact screens and commands needed to:

- create a normal IAM admin user
- create an AWS CLI access key for that user
- configure the AWS CLI in WSL
- launch the AWS FPGA Developer AMI with the repo launcher
- SSH into the instance and run the first Vitis HLS synthesis

Do not create root access keys. Use the root user only to get into the console and create your normal IAM user.

### Before You Start

You need:

- browser access to the AWS console
- root user access for the account, if this is the first setup pass
- WSL/Ubuntu on your workstation
- the repo cloned locally at `/mnt/c/Users/nomadic/projects/hngac-fpga`

On this machine the AWS CLI is already installed at `/usr/local/bin/aws`.

### Recommended Values

Use these values unless you have a reason to change them:

- IAM user name: `abuhassan-admin`
- AWS CLI profile: `hngac-aws`
- region: `us-east-1`

- repo launch target: f2-dev
- EC2 instance type: c7i.4xlarge
- root volume: 200 GiB gp3

## Screen 1: AWS Console Sign-In

1. Go to <https://console.aws.amazon.com/>.
2. Sign in as the **root user** if this is your first setup pass.
3. If AWS prompts you to set up MFA for the root user, do it now.

Stop here only if you do not have root access. Otherwise continue.

## Screen 2: IAM Users

1. In the AWS console, search for **IAM**.
2. Open **IAM**.
3. In the left nav, click **Users**.
4. Click **Create user**.
5. In the user name field, enter:

`abuhassan-admin`

6. If the console access toggle is shown and you want browser access too, enable it.
7. Continue to permissions.

## Screen 3: Permissions

1. Choose **Attach policies directly**.
2. Search for:

`AdministratorAccess`

3. Check **AdministratorAccess**.
4. Continue through review.
5. Click **Create user**.

You now have a normal IAM admin user for day-to-day work.

## Screen 4: Access Keys

1. Still in **IAM**, open **Users**.
2. Click the new user:

abuhassan-admin

3. Open the **Security credentials** tab.
4. Scroll to **Access keys**.
5. Click **Create access key**.
6. If the console asks for a use case, choose **Command Line Interface (CLI)**.
7. Continue.
8. Save both values immediately:
  - Access key ID
  - Secret access key

The secret is shown only once.

## Screen 5: AWS CLI In WSL

If AWS CLI is not already installed, use the official zip installer:

```
cd /tmp
curl -fsSL0 "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip"
unzip -q awscli-exe-linux-x86_64.zip
sudo ./aws/install
aws --version
```

If AWS CLI is already installed, skip to the next screen.

## Screen 6: Configure The Profile

In WSL, run:

```
aws configure --profile hngac-aws
```

When prompted, enter:

- AWS Access Key ID: paste the key you just created
- AWS Secret Access Key: paste the secret you just created
- Default region name: **us-east-1**
- Default output format: **json**

## Screen 7: Verify The Login

Run:

```
AWS_PROFILE=hngac-aws aws sts get-caller-identity
```

Expected result:

- an account ID

- an ARN for the IAM user
- no authentication error

If this fails, fix the CLI credentials before launching anything.

## Screen 8: Review The Repo Launch Preset

The repo includes a local launch preset at:

`scripts/aws/fpga-dev.env`

Open it if you want to check the defaults:

```
cd /mnt/c/Users/nomadic/projects/hngac-fpga
sed -n '1,220p' scripts/aws/fpga-dev.env
```

Current defaults:

- target: `f2-dev`
- instance type: `c7i.4xlarge`
- key pair name: `hngac-fpga-dev-abuhassan`
- PEM path: `/home/abuhassan/.ssh/hngac-fpga-dev-abuhassan.pem`
- owner tag: `abuhassan`

## Screen 9: Launch The Instance

From the repo root:

```
cd /mnt/c/Users/nomadic/projects/hngac-fpga
AWS_PROFILE=hngac-aws scripts/aws/launch_fpga_dev_instance.sh \
scripts/aws/fpga-dev.env
```

What the launcher does:

- checks AWS authentication
- finds the default VPC and subnet if you did not set custom ones
- creates or reuses a security group
- locks SSH to your current public IP
- creates the EC2 key pair if needed
- launches the instance with a `gp3` root volume
- waits for the instance to be ready
- prints the public IP, DNS, and SSH command

## Screen 10: SSH In

Use the SSH command printed by the launcher. It will look like:

```
ssh -i /home/abuhassan/.ssh/hngac-fpga-dev-abuhassan.pem \
    ubuntu@ec2-xx-xx-xx-xx.compute-1.amazonaws.com
```

## Screen 11: Verify Vitis

On the instance, run:

```
source /etc/profile.d/default_module.sh
vitis_hls -version
vivado -version
```

If those commands work, the AMI is ready for HLS work.

## Screen 12: Clone The Repo And Run HLS

On the EC2 instance, clone the repo onto the instance filesystem:

```
git clone <your-repo-url>
cd hngac-fpga
```

Then run the first HLS synthesis:

```
export HNGAC_HLS_PART=<your-target-part>
export HNGAC_HLS_CLOCK_NS=5.0
export HNGAC_HLS_WORKDIR=/tmp/hngac-fpga-hls
vitis_hls -f fpga/hls/scripts/vitis_hls.tcl
```

## If You Want A GUI

If you want Amazon DCV / GUI access later, set:

```
AWS_FPGA_ENABLE_DCV=true
```

in `scripts/aws/fpga-dev.env` before launching.

## F1 Is Off By Default

The launcher keeps the legacy F1 profile disabled.

Do not switch to:

```
AWS_FPGA_TARGET=f1-retired
AWS_FPGA_ENABLE_F1=true
```

unless you deliberately want the legacy path and know your account can still launch F1.

## Troubleshooting

### If AWS CLI login fails

Rerun:

```
AWS_PROFILE=hngac-aws aws sts get-caller-identity
```

If it still fails, recreate the access key or rerun `aws configure -profile hngac-aws`.

### If key creation fails

If the launcher says the key already exists, either:

- reuse it by setting `AWS_FPGA_CREATE_KEY_PAIR=false`
- or change `AWS_FPGA_KEY_NAME` in `scripts/aws/fpga-dev.env`

### If there is no default VPC

Set these explicitly in `scripts/aws/fpga-dev.env`:

```
AWS_FPGA_VPC_ID=<your-vpc-id>
```

```
AWS_FPGA_SUBNET_ID=<your-subnet-id>
```

## References

- AWS root user best practices: <https://docs.aws.amazon.com/IAM/latest/UserGuide/root-user-best-practices.html>
- Create an administrative user: <https://docs.aws.amazon.com/accounts/latest/reference/getting-started-step4.html>
- Create IAM users: [https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_users\\_create.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_users_create.html)
- Manage access keys: [https://docs.aws.amazon.com/IAM/latest/UserGuide/id\\_credentials\\_access-keys.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_access-keys.html)
- AWS CLI install guide: <https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>
- AWS FPGA Developer AMI docs: <https://awsdocs-fpga-f2.readthedocs-hosted.com/latest/User-Guide-AWS-EC2-FPGA-Development-Kit.html>
- AWS aws-fpga repository: <https://github.com/aws/aws-fpga>